Aarya Nema

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Technical Skills

Languages: Python, C++, JavaScript
Backend: Django, Node.js, Express.js, Electron
Frontend: React, TailwindCSS, HTML, CSS, Bootstrap
Clouds & Databases: AWS, PostgreSQL, MongoDB, FAISS
Web Technologies: Docker, Socket.IO, Redis
Developer Tools: Postman, VS Code, GitHub

Work Experience

Avyaan Management Limited (Bhopal) — **May 2026 – June 2026**
Software Development Intern | React.js, Express.js, MongoDB, Electron — *Offline*

- Built a full-featured **Point of Sale (POS)** system to support uninterrupted operations in offline environments, handling core sales workflows across multiple devices without continuous internet connectivity
- Developed an in-house **Finance Management App** to manage department-level financial workflows, incorporating role-based verification and partial/full payment handling for more controlled transaction processing
- Structured the application as a packaged, cross-platform solution with local data availability, enabling independent deployment across environments while maintaining consistent access to essential operational data

Amasqis.AI (Oman & Chennai) — **March 2025 – September 2025**
AI Intern | React.js, Node.js, REST APIs, Django, MongoDB — *Remote*

- Developed the backend foundation for a **Green Hydrogen Management System**, structuring the application around domain models, middleware, and route layers to support its operational data requirements
- Built data-ingestion workflows for multiple input sources, validating and normalizing incoming data before routing it through processing pipelines for downstream analysis and automated **report generation**
- Organized the platform around modular **REST APIs** with centralized request handling and input validation, making the backend easier to extend while limiting malformed data from reaching core workflows

Projects

AI Conversational Agent — *Ollama • Python • Silero VAD • React.js • WebSockets* — **Source Code**

- Built a real-time conversational agent to overcome the delays of traditional request-response voice systems, combining **Python**, **Ollama**, **Silero VAD**, **React.js**, and **WebSockets** into a continuous audio interaction pipeline
- Connected speech detection, agent processing, and custom Text-to-Speech generation through a streaming architecture, allowing audio to be processed and responses to begin generating without waiting for the entire interaction to complete
- Led development of the TTS and VAD components within a 2-person subteam, refining the streaming pipeline and reducing input-to-output latency by **80%** to deliver near real-time voice responsiveness

Adobe GenSolve – Semantic PDF Analyst — *Python • DistilBERT • FAISS • Docker* — **Source Code**

- Created a persona-driven document intelligence system for the **Adobe Hackathon (Round 1B)** to help users locate task-relevant information across large collections of PDF documents without manually searching through every file
- Built an **extract → process → rank** pipeline that converted PDF content into semantic representations with **DistilBERT** and used **FAISS** similarity search to surface the sections most relevant to a given persona and task
- Turned the retrieved results into structured JSON containing **top-k ranked sections and importance scores**, making the output directly usable for downstream analysis and automated document-processing workflows
- Packaged the complete pipeline with **Docker**, standardizing dependencies and runtime configuration so the system could be executed consistently across different environments with minimal setup

Certificates

ServiceNow System Administration Certificate — **2026**
Google IT Support Professional Certificate — **2025**

Achievements

Purplle Tech Challenge
Qualified for Round 2 by developing an AI-powered real-time CCTV analytics solution to track and analyze customer movement within retail stores.

Tata Steel AI Hackathon
*Ranked in the **Top 200** among **1000+** participants by developing a stacked ensemble machine learning model to predict optimal temperatures for steel rolls.*

Goldman Sachs Hackathon
*Scored **240+/300** by solving algorithmic and software engineering challenges involving data structures, optimization, and time-constrained problem-solving.*

Education

Vellore Institute of Technology, Bhopal — **2023 - 2027**
B.Tech in Computer Science & Engineering — *8.46/10 CGPA*